

SYLLABUS :-

The course emphasizes invaluable techniques and skills required to be an expert in organic synthesis. The following experiments will be conducted in this laboratory courses. (i) Special element detection by Na-fusion test (2x3h):Detection of elements like S, N, I, Cl, and Br from the known/unknown sample using Na fusion test.(ii) Functional group analysis (4x3h):Analysis of functional group like amine, nitro, alcohol, carbonyl and others from known and unknown solid/liquid samples.(iii) Filtration and recrystallization technique (3x3h):During this lab an impure sample of a known/unknown compounds will be provided. The task is to purify the sample by recrystallization and filtration, followed by reporting its melting point.(iv) Synthesis and Derivatization Melting point analysis (5x3h):The task is to derivatize the compounds and confirming them via melting point analysis. E.g. synthesis of aspirin from salicylic acid, 2,4-DNP hydrazone of cyclohexanone, Synthesis of benzillic acid from benzil. Book: Vogel, A. I. Elementary Practical Organic Chemistry: Qualitative Organic Analysis, Part-2, 2nd Ed, Pearson Education, 2010